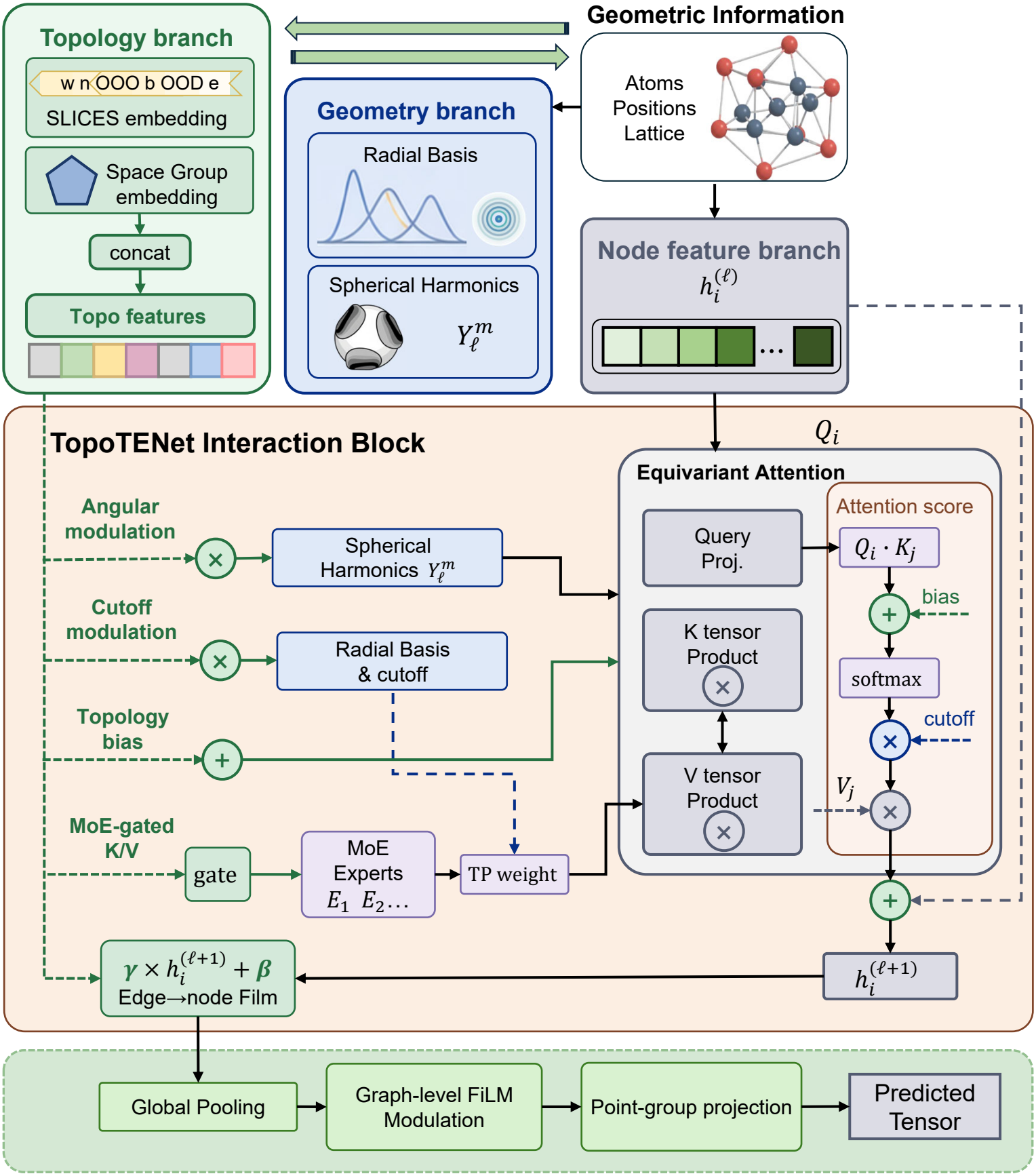
Topology branch
w n OOO b OOD e
SLICES embedding
Space Group embedding
concat
Topo features
Geometric Information
Atoms
Positions
Lattice
Geometry branch
Radial Basis
Spherical Harmonics
Y_ℓ^m
Node feature branch
$h_i^{(\ell)}$
TopoTENet Interaction Block
Q_i
Equivariant Attention
Angular modulation
Spherical Harmonics Y_ℓ^m
Cutoff modulation
Radial Basis & cutoff
Topology bias
MoE-gated K/V
gate
MoE Experts E_1 E_2...
TP weight
Query Proj.
K tensor Product
V tensor Product
Attention score
$Q_i \cdot K_j$
bias
softmax
cutoff
V_j
$\gamma \times h_i^{(\ell+1)} + \beta$
Edge→node Film
$h_i^{(\ell+1)}$
Global Pooling
Graph-level FiLM Modulation
Point-group projection
Predicted Tensor